Math 55 Second Midterm Exam, Prof. Srivastava
November 8, 2022, 5:10pm–6:30pm, 2050 VLSB.

Name: ______________________________

SID: ______________________________

GSI: ______________________________

Name of the student to your left: ______________________________
Name of the student to your right: ______________________________

Instructions: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.
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Sign here: ______________________________

Question	Points
1	10
2	7
3	7
4	7
5	9
6	10
Total:	50

Do not turn over this page until your instructor tells you to do so.

Name and SID: ____________________

1. (10 points) Circle true (**T**) or false (**F**) for each of the following. There is no need to provide an explanation. All graph theory definitions are provided on the last page of the exam.

   (a) If $n \geq 3$ and $1 < k < n$ then $\binom{n}{k}$ must be divisible by n. **T** **F**

   > **Solution:** This is false in general. For example if $n = 4$ and $k = 2$ then $\binom{4}{2} = 6$, which is not divisible by 4.

   (b) if m, n are positive integers and $m \leq n$ then for every $k \leq m$:

   $$\binom{m}{k} \leq \binom{n}{k}.$$

   T **F**

   > **Solution:** True. A combinatorial proof is: let A be a set with n objects and let B be a subset of A with m objects. Then $\binom{m}{k}$ is the number of k-subsets of B, which is at most the number of $k-$subsets of A (since every k-subset of B is a $k-$subset of A), which is $\binom{n}{k}$.

   (c) If a simple connected graph has an Eulerian circuit then it must be $3-$colorable.
   . **T** **F**

   > **Solution:** False. By a theorem in class, a graph has an Eulerian circuit if and only if every degree is even. The complete graph on 5 vertices has all degrees equal to 4, so it has an Eulerian circuit. However, it is not $3-$colorable since each vertex must be assigned a different color.

   (d) If a simple graph has an Eulerian circuit then it must contain at least two distinct spanning trees. **T** **F**

   > **Solution:** True. If a simple graph has an Eulerian circuit then every vertex must have even degree, which means it cannot be a tree (since it has no leaves). By the argument in Quiz 9 problem 1, this means it has at least three distinct spanning trees.

   [Scratch Space Below]

2. (7 points) Prove by induction that 5 divides $n^5 - n$ whenever n is a positive integer.

Solution: Let $P(n)$ be the statement that $5|n^5 - n$.

$P(1)$ is true since $n^5 - n = 1 - 1 = 0$ and 5 divides 0.

Assume $k \geq 1$ and assume $P(k)$. We will show $P(k+1)$. Observe that

$$(k+1)^5 - (k+1) = k^5 + \binom{5}{1}k^4 + \binom{5}{2}k^3 + \binom{5}{3}k^2 + \binom{5}{4}k + 1 - (k+1)$$

by the binomial theorem. By he induction hypothesis we have

$$5|k^5 - k.$$

Notice that $\binom{5}{1} = \binom{5}{4} = 5$ and $\binom{5}{2} = \binom{5}{3} = 10$ so 5 also divides

$$\binom{5}{1}k^4 + \binom{5}{2}k^3 + \binom{5}{3}k^2 + \binom{5}{4}k$$

. Thus, 5 must divide the sum of these expressions, which is equal to $(k+1)^5 - (k+1)$ by the first equation above, as desired.

3. (7 points) Consider the sequence of positive real numbers recursively defined by:

$$x_1 = 1, x_2 = 2, x_{n+1} = \sqrt{1 + x_n + x_{n-1}}.$$

Show using induction that $x_n \leq 4$ for all $n \geq 1$.

Solution: Let $P(n)$ be the statement that $x_n \leq 4$. We verify the base cases $P(1)$ ($x_1 = 1 \leq 4$) and $P(2)$ ($x_2 = 2 \leq 4$). For the inductive step, assume $n \geq 2$ and assume $P(1), \ldots, P(n)$. We will show $P(n+1)$. By the recursive definition,

$$x_{n+1} = \sqrt{1 + x_n + x_{n-1}}.$$

By the inductive hypothesis $x_n \leq 4$ and $x_{n-1} \leq 4$ so substituting and noting that the square root is an increasing function on $(0, \infty)$, we have

$$x_{n+1} = \sqrt{1 + x_n + x_{n-1}} \leq \sqrt{1 + 4 + 4} = \sqrt{9} = 3 \leq 4,$$

so $P(n+1)$ holds, as desired

4. (7 points) (a) How many subsets of $\{1, 2, \ldots, 17\}$ of size 5 contain at most one even integer? You can express your answer in terms of factorials (no need to simplify).

Solution: The given set contains 8 even integers $E = \{2, 4, 6, 8, 10, 12, 14, 16\}$ and 9 odd integers $O = \{1, 3, 5, 7, 9, 11, 13, 15, 17\}$. A subset with at most 1 even integer contains either 0 or 1 even integers. There are $\binom{9}{5}$ of the first kind of subset (we simply choose a subset of O) and $\binom{9}{4}\binom{8}{1}$ of the second kind by the product rule (we choose 4 elements of O and one element of E). Thus the total number of subsets with at most one even integer is

$$\binom{9}{5} + \binom{9}{4}\binom{8}{1}.$$

(b) How many subsets of $\{1, 2, \ldots, 17\}$ of size 5 contain at least one even integer?

Solution: Observe that if a is the number subsets of $\{1, \ldots, 17\}$ with 0 even integers and b is the number with at least one even integer then

$$\binom{17}{5} = a + b$$

since every subset has either 0 or at least one even inteer. We already calculated in the previous part that $a = \binom{9}{5}$. Thus the desired number is

$$b = \binom{17}{5} - \binom{9}{5}.$$

5. Recall that the degree sequence of a graph is a list of the degrees of its vertices in nondecreasing order. Give examples of graphs with the following properties, or prove that no such graph can exist.

   (a) (3 points) A simple graph with degree sequence $6, 5, 4, 3, 2, 1$.

   Solution: No such graph exists. By the handshaking theorem, the sum of the degrees of the vertices in a graph must be even. But the sum of the given degrees is $6(6+1)/2 = 21$, which is odd. You can alternatively observe that a simple graph with 6 vertices cannot have a vertex of degree 6 (since there are only 5 other vertices for it to be adjacent to).

   (b) (3 points) A $2-$colorable simple graph with degree sequence $4, 2, 2, 1, 1$.

   Solution: No such graph exists. Assume for contradiction that G is such a graph. Call the vertex of degree 4 x. Observe that x must connected to all of the other four vertices (since there are only 5 vertices total). One of these vertices, say y, must have degree 2 so it must be connected to another vertex in the graph, call it z. But now x, y, z form a circuit of length 3. Since a graph is $2-$colorable iff it contains no odd circuit, G cannot be $2-$colorable.

   (c) (3 points) A tree with degree sequence $3, 3, 2, 2, 1, 1$.

   Solution: No such tree exists. Observe that such a graph has 6 vertuces, and the sum of the degrees is 12, so it has 6 edges by the handshaking theorem. But we proved in class that a tree on n vertices has $n-1$ edges, so this graph cannot be a tree.

6. (a) (3 points) Prove that if $n \geq 2$ and x, y are positive integers such that $x + y = n$ then

$$\binom{x-1}{2} + \binom{y-1}{2} \leq \binom{n-1}{2}.$$

Solution: *This question turned out to be longer than I indended. 6a was meant to be used as a step in 6b but I made a mistake and wrote* $\binom{x-1}{2} + \binom{y-1}{2}$ *instead of* $\binom{x}{2} + \binom{y}{2}$ *so it is not directly relevant to 6b, though it does serve as a good hint. Nonetheless 6a is a true statement. It is possible to prove it algebraically or combinatorially.*

*Algebraic Proof.*If $x \leq 2$ and $y \leq 2$ then the left hand side is zero so the inequality is true. If $x \leq 2$ and $y \geq 3$ then $\binom{x-1}{2} = 0$ so the inequality becomes $\binom{y-1}{2} \leq \binom{n-1}{2}$, which is true by Question 1b. for an analogous reason the inequality holds when $y \leq 2$ and $x \geq 3$.

Otherwise, the desired inequality reads

$$(x-1)(x-2)/2 + (y-1)(y-2)/2 \leq (n-1)(n-2)/2$$

which after cancelling 2s and expanding is

$$x^2 - 3x + 2 + y^2 - 3y + 2 \leq n^2 - 3n + 2$$

which substituting $n = x + y$ and cancelling $2's$ and $-3x - 3y = -3n$ from both sides is

$$x^2 + y^2 + 2 \leq (x+y)^2$$

which is true since $(x+y)^2 - 2 = x^2 + y^2 + 2xy - 2 \geq x^2 + y^2$ since $x, y \geq 2$.

Combinatorial Proof. By the argument in the algebraic proof above, we may assume $x, y \geq 3$. If we let $A = \{1, \ldots, n-1\}$ and consider the sets $B = \{1, \ldots, x-1\}$ $C = \{x+1, \ldots, x+(y-1\}$ then $B \subseteq A$ and $C \subseteq A$ since $x + y - 1 = n - 1$, and $B \cap C = \emptyset$. Observe that

$$\binom{n-1}{2} = |\{S \subseteq A : |S| = 2\}|.$$

and

$$\binom{x-1}{2} + \binom{y-1}{2} = |\{S \subseteq A : |S| = 2, S \subseteq B \vee S \subseteq C\}|.$$

Since

$$\{S \subseteq A : |S| = 2, S \subseteq B \vee S \subseteq C\} \subseteq \{S \subseteq A : |S| = 2\},$$

the desired inequality follows.

(b) (7 points) Prove that if G is a simple graph with $n \geq 1$ vertices and at least $\binom{n-1}{2}+1$ edges, then G must be connected.

Solution: If $n = 1$ then there is nothing to show since there is no simple graph with one vertex and at least 1 edge. If $n = 2$ then there is a unique simple graph with at least 1 edge, namely the graph with one edge and two vertices, which is connected.

Assume $n \geq 3$. We will show the contrapositive. Assume G is disconnected. We will show that G has at most $\binom{n-1}{2}$ edges. Let $H_1, \ldots, H_k$ be the connected components of G. Let $A = H_1$ and $B = H_2 \cup \ldots \cup H_c$ and notice that there are no edges between A and B, since otherwise we would be able to add this edge to A to make it a larger connected component.

Let x be the number of vertices in A and let y be the number of vertices in B, and notice that $x + y = n$. The number of pairs of vertices in A is $\binom{x}{2}$ so it has at most $\binom{x}{2}$ edges. Similarly B has at most $\binom{y}{2}$ edges.

The desired conclusion is implied by the following claim (which is a bit stronger than 6a) since all edges of G must lie in A or B:

Claim. $\binom{x}{2} + \binom{y}{2} \leq \binom{n-1}{2}$.

*Proof of Claim.*If $x = 1$ then $y = n - 1$ and we have $\binom{x}{2} + \binom{y}{2} = 0 + \binom{n-1}{2}$ so the claim holds. The same argument applies when $y = 1$.

So assume $x, y \geq 2$. Expanding the definition of the binomial coefficient, the inequality we want to prove is

$$x(x-1)/2 + y(y-1)/2 \leq (n-1)(n-2)/2,$$

which cancelling the 2's is

$$x^2 + y^2 - (x + y) \leq n^2 - 3n + 2.$$

Substituting $x + y = n$ and rearranging this becomes

$$x^2 + y^2 \leq (x + y)^2 - 2n + 2 = x^2 + y^2 + 2xy - 2x - 2y + 2,$$

which on cancelling $x^2 + y^2$ from both sides is:

$$2x + 2y \leq 2xy + 2$$

or equivalently

$$x + y \leq xy + 1.$$

This is implied by $x + y \leq xy$ which is true since $x/xy + y/xy = 1/y + 1/x \leq 1/2 + 1/2 \leq 1$ as $x, y \geq 2$, completing the proof.

Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of 2−element subsets of V, called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y. The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \ldots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \ldots k$. The path is said to **connect** x_0, x_k, **visit** the vertices $x_0, \ldots, x_k$, and **traverse** the edges $e_1, \ldots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k-coloring** of G is a function $f : V \to \{1, \ldots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y. A graph is $k-$**colorable** if it has a k-coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G. A **connected component** of G is a maximal connected subgraph of G, i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the endpoints. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.